Camp Search Concierge

CSC Plan Guide — Updated

OurKids.net | Version 2.0 | March 2026

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SECTION 1

CSC Architecture

Hybrid Search Engine

Use MySQL v8 for structured queries and Google Gemini for semantic reasoning. The system uses Fuzzy Matching and vector-based RAG to determine a Relevance Matching Percentage (RMP) score.

Intelligent Fallback Logic

- CSSL (Camp SQL Search Logic): Primary search engine for high-confidence direct matches. Executes FULLTEXT and FUZZY queries against the programs and activity_tags tables.
- CASL (Camp AI Semantic Logic): Triggered when the RMP score falls below a defined threshold. Uses Google Gemini with the activity taxonomy and program descriptions as RAG context to generate a hybrid RMP score.

Conversational Guardrails & Clarification

If inputs are ambiguous or RMP scores remain low after both CSSL and CASL, the CSC engages the user with proactive discovery questions based on the camp data model (e.g. 'Are you looking for a day camp or overnight?'). The agent maintains a polite persona and includes safety filters to block inappropriate content.

Optimization & Cost Control

Architect CASL to use the most cost-effective Gemini models to minimize token consumption and latency for high-volume traffic (1,000+ daily users). The sitems.related_ids field provides a pre-built semantic expansion graph — enabling CASL to broaden tag queries without a Gemini call for known tag relationships, significantly reducing API costs.

Discovery Features

After a search, the CSC will offer a "Surprise Me" curated list of 10 recommendations based on trends and popularity, with Gold Tier camps receiving priority placement.

Legacy Integration

Use the campsearchv2 project directory (category taxonomy, baseline logic) as the foundation. The legacy sitems table (392 activity tags with hierarchy, related-tag graph, and color coding) and the existing tier assignments in camps.eListingType provide the core data structures that the new schema is built on.

SECTION 2 Display & Ranking Logic

Result Format

Search results must be delivered with high speed and follow a strict visual hierarchy:

- Line 1: Program Name (from programs.name)
- Line 2: Camp Name (from camps.camp_name)
- Line 3: Context — a personalized blurb explaining the specific RMP match and why it fits the search. Source: programs.mini_description, enriched by CASL when needed.
- Line 4: Cost (from programs.cost_from / cost_to)
- Line 5: Location (from camps.city, camps.province)

Tiered Priority

Ranking is determined by RMP score, but Gold Tier camps that meet the minimum threshold must be pinned to the top of results. Silver and Bronze tier camps rank by RMP below them.

Deduplication

If a single camp has multiple programs with high RMP scores, display all relevant program entries — do not collapse them. Each program is a distinct searchable offering.

Filtering

Only display results that meet the minimum input requirements. Suppress irrelevant data. Apply hard filters (age, location, type) before RMP ranking, not after.

SECTION 3

Data Foundation

★ *New Section — Added based on legacy database analysis*

The CSC database is built from three source documents and one legacy database backup, all of which have been fully analyzed. This section defines what each source contributes to the new schema and how they connect.

3.1 Source Documents & Their Roles

Source	Type	Role in CSC
ALL_CAMP_PAGES (735 URLs)	Category pages	Defines the navigation/SEO landing pages. Each row is a filtered page with SQL logic. Populates the categories table and informs CASL's RAG context for semantic page matching.
Activity Taxonomy PDF	Tag hierarchy	Master reference for the activity_tags table. 6 domains → 114 categories → 111 sub-activities = 225 canonical tags. Validated against and supersedes the legacy webItems table.
AI Quiz Requirements PDF	UI/logic spec	Defines the conversational search quiz interface and the clarification question logic used when CSSL and CASL both return low RMP scores.
camp_directory DB dump	Legacy database	~2,100 camps, ~24,260 programs, 392 activity tags (sitems), 12 character traits, full tier assignments, Lat/Lon, cost/age data. The primary ETL source for the new schema.

3.2 Activity Taxonomy — The RAG Source

The activity taxonomy is the most critical data asset for both CSSL (tag-based filtering) and CASL (semantic reasoning). It exists in two forms in the legacy system — the newer sitems table is the authoritative source:

Feature	Legacy webItems (old)	sitems → activity_tags (new)
Total tags	171	392
Hierarchy depth	2 levels	3 levels (Domain → Category → Sub-activity)
Semantic graph	None	Yes — related_ids field (pre-built tag expansion, reduces Gemini API calls)
Slug / short name	No	Yes — URL slugs + short_name for UI display
Color coding	No	Yes — per domain, for UI

Used for CSC	Legacy SQL bridge only	CSSL filtering + CASL RAG + category page mapping

The 6 top-level domains and their tag counts in the new taxonomy:

Domain	Domain	Domain	Domain
Adventure7 categories, 0 sub-activities	Arts16 categories, 51 sub-activities	Computers & Tech20 categories, 8 sub-activities	Education29 categories, 18 sub-activities
Health & Fitness11 categories, 0 sub-activities	Sports31 categories, 34 sub-activities	Total: 225 tags across 114 categories	Source: sitems table (392 entries incl. special needs, religious, labels)

3.3 Legacy Database — Key Findings

Analysis of camp_directory_dump20260205.sql (54 tables, 64MB uncompressed) produced the following findings relevant to the new schema:

- ~2,100 active camps with tier assignments (gold/silver/bronze) already set in camps.eListingType — no re-classification needed, direct migration.
- ~24,260 program sessions across all camps. This is the primary searchable unit for CSSL queries.
- Lat/Lon fields exist in the camps table but are stored as VARCHAR — must be retyped to DECIMAL(10,7) in the new schema for spatial indexing.
- The sitems table (392 tags) includes a relatives field — a comma-separated list of semantically related tag IDs. This is a pre-built semantic expansion graph that CASL can use to broaden queries without a Gemini call, directly reducing API costs at scale.
- A fully built character trait system exists: 12 traits (Resilience, Curiosity, Courage, Independence, etc.) linked to programs via a junction table with justification text. This maps to the character-trait category pages in the URL list and should be surfaced as a CSC filter dimension.
- programs.mini_description (500 chars) is the ideal source for the CSC Line 3 context blurb. programs.description (full text) feeds CASL's RAG pipeline.
- ~20 legacy tables (print ads, InDesign codes, COVID flags, media galleries, staging drafts) should be dropped during ETL — they have no search value.

3.4 New Database Schema — Core Tables

The following 6 tables form the complete data model for the CSC. All legacy data maps into this structure via the ETL process defined in Section 4.

camps — Core camp entity

Field	Type	Notes

id	INT PK AUTO_INCREMENT	Replaces cid (smallint too small)
camp_name	VARCHAR(150)	FULLTEXT indexed
slug	VARCHAR(200)	URL identifier
tier	ENUM('gold','silver','bronze')	Migrated from eListingType. Indexed for ranking.
status	TINYINT	1=active, 0=inactive
lat	DECIMAL(10,7)	Fixed: was VARCHAR in legacy
lon	DECIMAL(10,7)	Fixed: was VARCHAR in legacy
city	VARCHAR(80)	Denormalized from addresses for query speed
province	VARCHAR(60)	
country	TINYINT	1=Canada, 2=USA, 3=International
website	VARCHAR(200)	
description	TEXT	FULLTEXT indexed. CASL RAG context.
mission	VARCHAR(300)	From generalInfo_web.mission
lgbtq_welcoming	TINYINT	lgbtq1 + lgbtq2 from generalInfo_web
accessibility	TINYINT	Accessibility support flag
review_count	SMALLINT	Denormalized for ranking speed
review_avg	FLOAT	Denormalized for ranking speed

programs — The searchable unit (replaces sessions)

Field	Type	Notes
id	INT PK AUTO_INCREMENT	
camp_id	INT FK → camps.id	Indexed
name	VARCHAR(300)	Program name. FULLTEXT indexed. → CSC Line 1
type	VARCHAR(45)	Day / Overnight / Class / Virtual / PA Day
start_date	DATE	Indexed
end_date	DATE	Indexed
age_from	SMALLINT	Indexed for age-range filtering
age_to	SMALLINT	Indexed
cost_from	SMALLINT	In CAD. → CSC Line 4
cost_to	SMALLINT	In CAD. → CSC Line 4
gender	TINYINT	0=Coed, 1=Boys, 2=Girls

is_special_needs	TINYINT	
is_virtual	TINYINT	
is_family	TINYINT	
language_immersion	VARCHAR(45)	ESL/FSL codes from legacy
tagline	VARCHAR(100)	Short display label
mini_description	VARCHAR(500)	→ CSC Line 3 context blurb source
description	TEXT	Full text. FULLTEXT indexed. → CASL RAG
status	TINYINT	Replaces draft tables. 1=live, 0=draft

activity_tags — Master taxonomy (replaces sitems)

Field	Type	Notes
id	INT PK	Preserve legacy sitems.id for migration continuity
parent_id	INT FK → self	NULL for top-level domain entries
domain_id	INT FK → activity_tags	Shortcut to L1 domain for fast filtering
name	VARCHAR(150)	Display name. FULLTEXT indexed.
short_name	VARCHAR(125)	UI label from sitems.short_name
slug	VARCHAR(150)	URL-safe identifier for category page mapping
level	TINYINT	1=Domain, 2=Category, 3=Sub-activity
tag_type	VARCHAR(20)	specialty / specialneeds / religious / label
related_ids	TEXT	From sitems.relatives — semantic expansion graph for CASL
color_code	VARCHAR(10)	Hex color per domain for UI
is_active	TINYINT	Hide deprecated tags without deleting

program_tags — Junction: programs ↔ activity_tags

Field	Type	Notes
program_id	INT FK → programs.id	Composite PK with tag_id. Indexed.
tag_id	INT FK → activity_tags.id	Indexed
is_primary	TINYINT	1 = primary tag (from sessions.specialty). Secondary from sessions.specialty2

categories — SEO/navigation landing pages (from ALL_CAMP_PAGES)

Field	Type	Notes
id	INT PK	From ALL_CAMP_PAGES ID column
title	VARCHAR(200)	Page title
meta_title	VARCHAR(200)	SEO meta title
slug	VARCHAR(250)	Full URL path
filter_type	VARCHAR(100)	SQLtype value (Sports, Arts, Education, etc.)
filter_activity_tags	TEXT	Comma-separated activity_tags.slug values (replaces SQLwebitems)
filter_city	TEXT	City list for geo-filtered pages
filter_province	VARCHAR(100)	
filter_country	TINYINT	
filter_gender	VARCHAR(20)	
filter_religion	VARCHAR(50)	
filter_day_overnight	VARCHAR(100)	
filter_options_sql	TEXT	Raw legacy SQL for complex filters (age, date, cost, ESL, traits, LGBTQ)
is_active	TINYINT	

program_traits + traits — Character trait system (migrate as-is)

Field	Type	Notes
traits.id	INT PK	12 traits: Resilience, Curiosity, Courage, Independence, etc.
traits.name	VARCHAR(150)	Display name
traits.slug	VARCHAR(150)	Maps to category page slugs (e.g. /camp/resilience)
program_traits.program_id	INT FK → programs.id	
program_traits.trait_id	INT FK → traits.id	
program_traits.justification	VARCHAR(500)	Why this program builds this trait — CASL context

3.5 How the Data Connects to CSSL & CASL

The following diagram shows the data flow from user query to result:

CSSL — SQL Layer	CASL — AI Layer

Queries against: ○ programs (FULLTEXT on name, description) ○ activity_tags (tag slug matching) ○ program_tags (junction lookup) ○ camps (tier, city, province, lat/lon) **Hard filters applied first:** ○ Age range ○ Location / Province ○ Program type (Day/Overnight) ○ Cost range ○ Gender	**RAG context sources:** ○ activity_tags.name + related_ids (tag graph) ○ programs.description (full text) ○ program_traits.justification (trait context) ○ session_activities.activities (free-text) ○ categories (landing page descriptions) **Cost optimization:** *Use related_ids tag expansion before calling Gemini. Only invoke Gemini API when SQL + tag graph both fail to reach RMP threshold.*

3.6 Character Traits — Hidden Asset

The legacy database contains a fully built character trait system that is largely unexposed to users today. It consists of 12 traits — Self-regulation, Independence, Responsibility, Courage, Resilience, Interpersonal Skills, Curiosity, Creativity, Physicality, Generosity, Tolerance, and Religious Faith — each linked to specific programs with a written justification explaining why that program builds the trait.

This system maps directly to 12 existing category pages in the URL list (e.g. /camp/resilience, /camp/curiosity, /camp/courage). For the CSC, these traits should be surfaced as a filter dimension — particularly valuable for parents searching with developmental goals in mind rather than activity names. It is a strong differentiator versus other camp search tools.

SECTION 4

Data Modeling & Migration

Schema Modernization

Design a lean, high-performance database model focused on the essential fields required by the CSC. The 6-table schema defined in Section 3.4 replaces ~54 legacy tables, retaining only the data that drives search, ranking, display, and AI reasoning.

ETL & Migration — Phase Order

Migration must follow dependency order. Each phase must be complete and validated before the next begins:

Phase	Tables	Key Actions
1	**activity_tags(from sitems)**	Foundation — all other tables reference it. Migrate sitems → activity_tags. Generate slugs. Map h1/h2/h3 → level field. Preserve related_ids (relatives). Validate all 392 tags against Activity Taxonomy PDF.
2	**camps + addresses**	Migrate camps + addresses. Denormalize city/province/postal into camps. Retype Lat/Lon from VARCHAR → DECIMAL(10,7). Map eListingType → tier enum. Verify all ~2,100 camp records.
3	**programs(from sessions)**	Migrate sessions → programs. Clean draft rows (status=0). Map type values to new enum. Verify age_from/age_to. Preserve mini_description and description for search and CASL.
4	**program_tags**	Build junction from sessions.specialty + sessions.specialty2 → program_tags rows. Set is_primary=1 for specialty, 0 for specialty2. Validate all tag IDs exist in activity_tags.
5	**categories(from ALL_CAMP_PAGES)**	Parse 735 URL rows. Map SQLwebitems values → activity_tags.slug references. Store complex SQL options (ESL, trait, LGBTQ, date filters) in filter_options_sql for now.
6	**traits + program_traits**	Migrate as-is from legacy trait and program_trait tables. Add slug field to traits (map to category page URLs). Low risk, high value.

Data Cleansing

- Strip ~20 legacy tables with no search value: ad_book, brand_identity, camps_2013, covid, alumni, stories, reflections, photoGallery, okp_gallery, ps_photos, media, camp_proofs, impressions, follow, sessions_draft, session_date_draft, location_tmp, promos, videos (separate to object storage).
- Normalize city names — legacy data has inconsistent casing (e.g. 'toronto' vs 'Toronto' vs 'north york').
- Convert Lat/Lon to DECIMAL(10,7) — mandatory for spatial index and proximity search.

- Remove or flag sessions with missing specialty (tag) — these cannot be matched by CSSL.
- Map legacy webItems references in category SQL filters to new activity_tags slugs.

Tier Mapping

The Gold/Silver/Bronze tier assignments are already present in camps.eListingType. Migration steps:

- Direct value copy: 'gold' → 'gold', 'silver' → 'silver', 'bronze' → 'bronze'.
- Verify ~2,100 records — flag any NULL or unrecognized values for manual review.
- Cross-reference with generalInfo_web.eListingType — use camps value as authoritative, flag mismatches.

Geo-Data

Ensure Latitude and Longitude are correctly indexed for proximity searches:

- Retype camps.Lat and camps.Lon from VARCHAR(100) to DECIMAL(10,7).
- Add spatial index: CREATE SPATIAL INDEX idx_geo ON camps (lat, lon) — or use a generated POINT column for MySQL 8 ST_Distance queries.
- Validate non-null lat/lon for all active Gold Tier camps as a priority — these are pinned to top of results.
- For camps with missing geo-data, geocode from city/province/postal using a one-time geocoding script.

SECTION 5

Tech Stack

Layer	Technology	Role
AI / Semantic	Google Gemini	CASL semantic reasoning. Most cost-effective model tier for high-volume queries. Tag graph (related_ids) used first to reduce Gemini calls.
Platform / UI	Streamlit	Search interface, filter controls, results display, clarification loop UI, Surprise Me feature.
Database	MySQL 8	CSSL FULLTEXT and structured queries. Spatial index for proximity search. InnoDB engine for all new tables.
Code / ETL	Python	Search orchestrator, ETL migration scripts, RMP scoring logic, CASL API wrapper, data cleansing.
Object Storage	TBD	Camp photos, logos, gallery images — decoupled from MySQL.

SECTION 6

Search Logic Flow

The CSC follows a tiered Waterfall approach — starting with the most efficient method and escalating only when necessary.

Step 1 — User Input Analysis

The Streamlit interface captures the search query. The Search Orchestrator (Python) checks whether this is a new search or a follow-up refinement within the same session.

Step 2 — CSSL: Camp SQL Search Logic

- Execute a MySQL FULLTEXT or FUZZY query against programs.name, programs.description, and activity_tags.name.
- Apply hard filters: age, location/province, program type, cost range, gender.
- JOIN program_tags → activity_tags to match tag slugs from the query.
- Calculate initial RMP score based on field match weight (tag match = high weight, FULLTEXT = medium, fuzzy = lower).
- Decision: If RMP ≥ threshold (e.g. >85%), bypass CASL and return results immediately.

Step 3 — CASL: Camp AI Semantic Logic

- Triggered only when CSSL RMP falls below threshold.
- First pass: Use activity_tags.related_ids to expand the query to semantically related tags. Re-run CSSL with expanded tag set. Recalculate RMP.
- Second pass (if still below threshold): Call Gemini API with the user query + RAG context (taxonomy, program descriptions, category page data). Gemini returns a refined tag set and reasoning.
- Assign Hybrid RMP score combining SQL match confidence and Gemini semantic confidence.
- Re-rank results using Hybrid RMP. Apply Gold Tier pinning.

Step 4 — Threshold Check & Clarification Loop

- If both CSSL and CASL fail to meet the minimum accuracy threshold, trigger the Clarification Loop.
- The CSC asks a targeted question to refine parameters, based on what dimensions are missing from the query.
- Example questions: 'Are you looking for a day camp or overnight?', 'What age is your child?', 'Do you have a city or region in mind?', 'Are you interested in a specific activity or something more general?'
- User response feeds back into Step 2 as a refined query.

Step 5 — Result Display

- Rank all results by RMP score.
- Pin Gold Tier camps meeting the threshold to the top, ordered by RMP within the Gold group.

- Format each result: Line 1 (Program Name) → Line 2 (Camp Name) → Line 3 (Context blurb) → Line 4 (Cost) → Line 5 (Location).
- Deduplicate: show all high-RMP programs from the same camp as separate entries.
- Suppress results below minimum RMP threshold.

Step 6 — Surprise Me

After results are displayed, offer a 'Surprise Me' prompt. Returns a curated list of 10 camps based on trending searches and popularity scores, with Gold Tier camps guaranteed visibility regardless of exact query match. This feature runs a separate lightweight query against cached popularity data — it does not invoke CASL.

RMP Scoring Reference

Score Range	Status	Action
> 85%	High confidence	CSSL match — bypass CASL, display results immediately
70–85%	Medium confidence	Display results, optionally trigger CASL for enrichment
< 70%	Low confidence	Trigger CASL — tag graph expansion first, then Gemini
< 50% (post-CASL)	Very low	Trigger Clarification Loop — ask user a refinement question